Fragments (subgraphs) in complex networks

4

> Beauty: the adjustment of all parts proportionately so that one cannot add or subtract or change without impairing the harmony of the whole.
>
> L. Battista Alberti

Informally, a subgraph is a part of a network. The total number of subgraphs of a given type is an appropriate measure used to characterise a network as a whole. For instance, the number of nodes and links, which are the simplest subgraphs, are global topological characteristics of complex networks.

Formally, a subgraph is a graph $G' = (V', E')$ of the network $G = (V, E)$ if, and only if, $V' \subseteq V$ and $E' \subseteq E$. If the graph G is a digraph, the corresponding subgraphs are directed subgraphs. There are different types of subgraph, as illustrated in Figure 4.1. In general, a subgraph can be formed by one (connected subgraph) or more (disconnected subgraphs) connected components. They can be acyclic or cyclic, depending on the presence or absence of cycles in their structures.

The use of subgraphs to characterise the structure of networks dates back to 1964, when the Russian chemist E. A. Smolenskii (1964) developed a substructural molecular approach to quantitatively describe structure–property relationships. Smolenskii's procedure is based on the decomposition of the molecular graph into different fragments or subgraphs. The subgraphs of a given type then become a variable in a multiple regression model to describe a physical property of molecules. This idea was further developed in the 1980s as the *graph-theoretic cluster expansion ansatz* (Klein, 1986; McHughes and Poshusta, 1990). In general, these methods express a molecular property P as a sum of contributions of all connected subgraphs of the network G:

$$P(G) = \sum_k \eta_k(G) \cdot N_k \tag{4.1}$$

where η is the contribution of the k th subgraph to the molecular property, and the *embedding frequencies* N_k are the number of times a given subgraph appears in the molecular graph (Klein, 1986). The sum in (4.1) extends over all connected subgraphs in the network. A variation of the substructure descriptors used in chemistry is *molecular fingerprints* (Willett, 2006)—string representations of the chemical structure in which every entry of the string represents

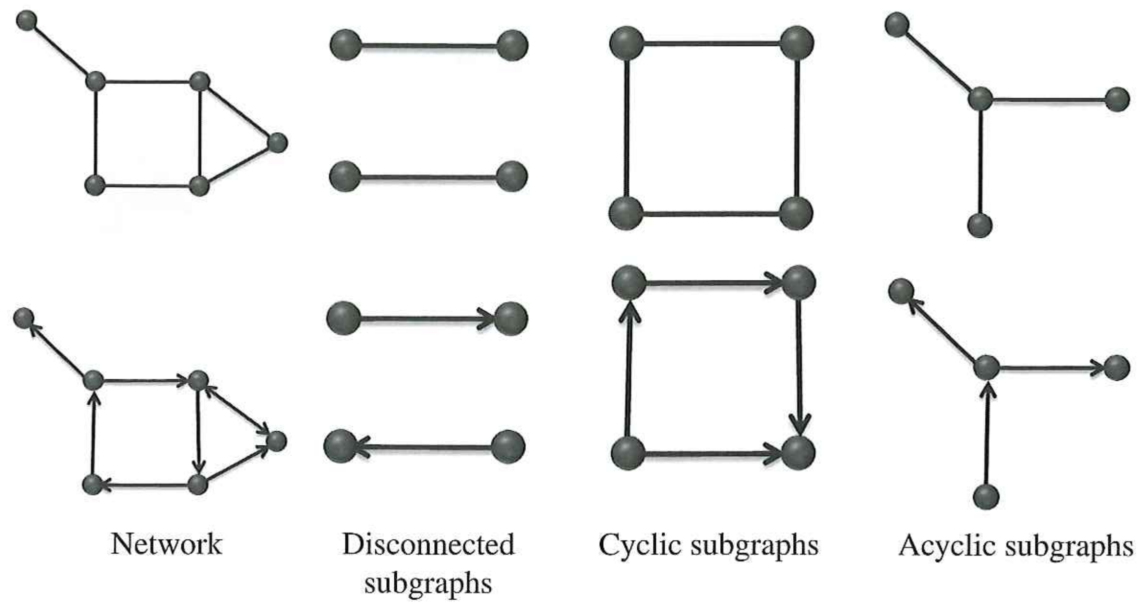

Fig. 4.1
Subgraphs. Subgraphs in undirected (top) and directed (bottom) networks.

a subgraph descriptor associated with a specific molecular feature. Molecular fingerprints are widely used in similarity searching, in which a series of molecules in a database is compared with a target molecular structure—usually with a desired property—allowing a ranking of decreasing similarity with the target for all the molecules. The molecular fingerprints are used as a vector characterizing every structure, and are compared to others by using an appropriate distance function, such as the Euclidean distance:

$$d_{ij} = \sqrt{\sum_{r=1}^{p} (f_{ir} - f_{jr})^2} \tag{4.2}$$

where f_{ir} is rth entry of the molecular fingerprint for the target molecule, and f_{jr} is the same entry for the jth molecule in the database.

4.1 Network 'graphlets'

The idea of molecular fingerprints has been applied to protein–protein interaction networks under the name of 'graphlets' (Pržulj et al., 2004; Pržulj, 2007). A *graphlet* is a rooted subgraph in which nodes are distinguished by their topological equivalence. For instance, in a path of length 2 there are two kinds of node from a topological point of view: the node in the middle and the two (equivalent) end nodes. Pržulj has defined 73 non-equivalent types of node which are present in the 30 subgraphs having from 2 to 5 nodes (Pržulj et al., 2004; Pržulj, 2007). In Figure 4.2 we illustrate these graphlets

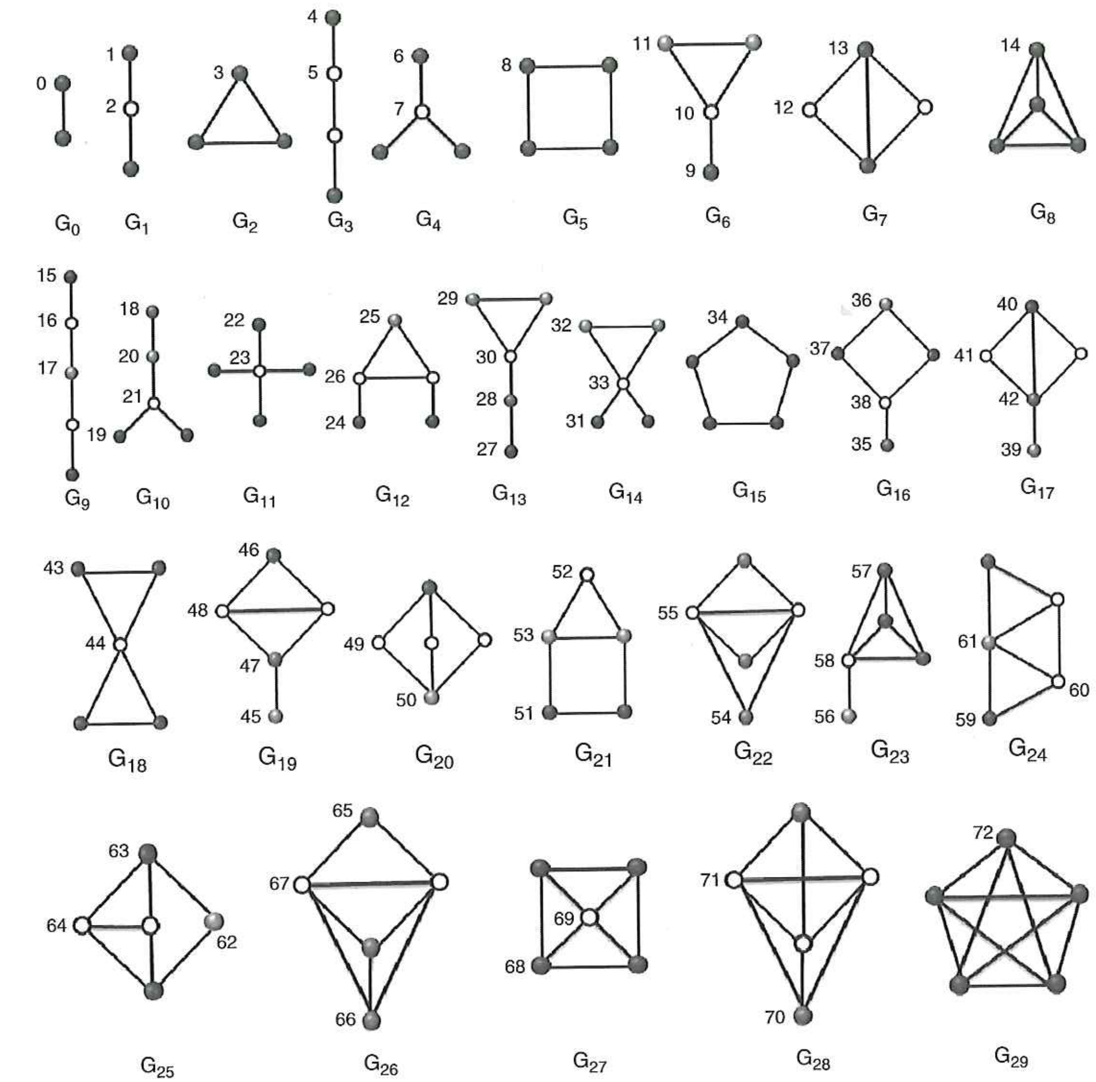

Fig. 4.2
Graphlets. Graphlets are rooted subgraphs having from three to five nodes. Here, nodes which are in the same automorphism orbit in a graphlet are drawn identically: dark grey, light grey, or empty circles.

in which non-equivalent nodes are numbered from 0 to 72. Then, the graphlet is a network fingerprint formed by a vector having 73 entries. Every entry of the graphlet represents the number of nodes of the corresponding type which are present in a given network.

Fragments, fingerprints, and graphlets can be used in different ways to study complex networks. Fragments have been the basis of quantitative structure–property relationships, such as the cluster expansion method. In this method, some properties are correlated with the number of fragments of different types in a group of networks. Another area of intensive research is the study of (molecular) similarity (Willett, 1987). Here the vector of (molecular) fingerprints of a series of objects is used to define a similarity function between the corresponding objects (molecules, networks), which are classified in different groups. Graphlets, on the other hand, have been used to extend the idea of degree distributions to subgraphs, as well as for finding similarities between proteins in a protein–protein interaction network. Some of these examples of applications will be analyzed in the second part of this book.

4.2 Network motifs

Another application of the number of subgraphs in complex networks is the concept of *network motif* (Milo et al., 2002; 2004a; 2004b; Artzy-Randrup et al., 2004). This concept was introduced by Milo et al. (2002) in order to characterize recurring significant patterns in real-world networks. A network motif is a subgraph that appears more frequently in a real network than could be expected if the network were built by a random process. This means that in order to find a motif in a real-world network we need to compare the number of occurrences of a given subgraph in this network with the number of times this subgraph appears in the ensemble of random networks with the same degree sequence. The appearance of a motif in a network then depends on the random model we select for comparison. Consequently, a large number of random networks is necessary to compute the frequency of appearance of the subgraph under study. A subgraph is then considered a network motif if the probability P of appearing in a random network an equal or greater number of times than in the real-world network is lower than a certain cut-off value, which is generally taken to be $P = 0.01$.

The statistical significance of a given motif is calculated by using the *Z-score*, which is defined as follows for a given subgraph i:

$$Z_i = \frac{N_i^{real} - \langle N_i^{random} \rangle}{\sigma_i^{random}} \tag{4.3}$$

where N_i^{real} is the number of times the subgraph i appears in the real network, $\langle N_i^{random} \rangle$, and σ_i^{random} are the average and standard deviation of the number of times that i appears in the ensemble of random networks, respectively. In Figure 4.3 we illustrate some of the motifs found by Milo et al. (2002) in different real-world directed networks.

A characteristic feature of network motifs is that they are network-specific. That is, a motif for a given network is not necessarily a motif for another network. However, a family of networks can be characterized by having the same series of motifs. In order to characterize such families, Milo et al. (2004a) have proposed using a vector whose ith entry produces the importance of the ith motif with respect to the other motifs in the network. Mathematically, the *significance profile* vector is expressed as

$$SP_i = \frac{Z_i}{\sum_j Z_j^2} \tag{4.4}$$

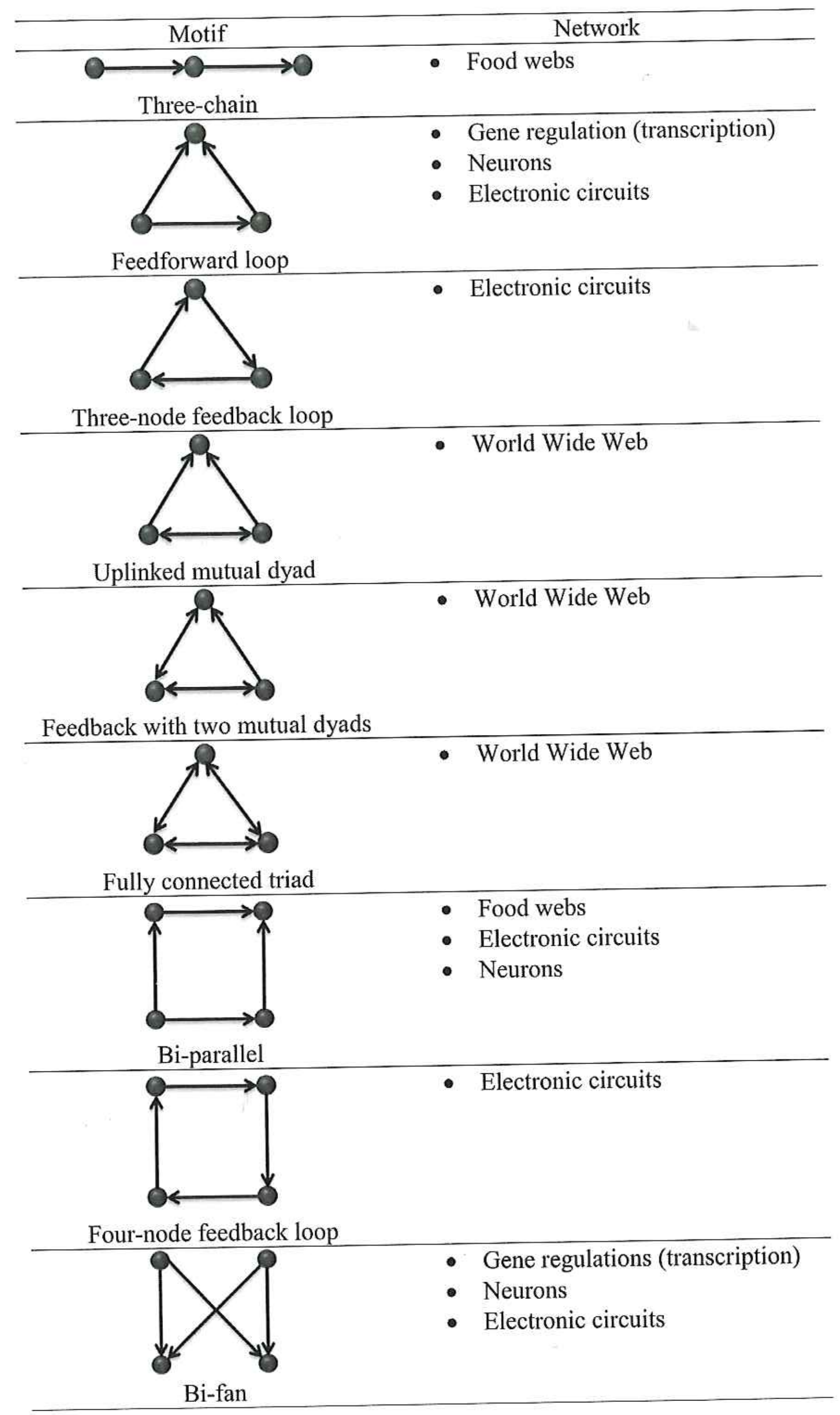

Fig. 4.3
Network motifs. Some of the motifs found in real-world networks, according to Milo et al. (2002).

4.3 Closed walks and subgraphs

Closed walks are very ubiquitous in our everyday life. If we begin our daily journey from home to our office and then return home, we have completed a closed walk of length 2. Sometimes after working we visit the supermarket and end the day by returning home. On those days we complete a closed walk of length 3. In general, a closed walk is a sequence of (not necessarily different) nodes and links in the network, beginning and ending at the same node.[1] By continuing with this analogy we can represent the places and streets we have visited at least once during a closed walk as nodes and links, respectively. In doing so, we are associating a subgraph with every closed walk. For instance, a closed walk of length 2 is associated with a subgraph formed by two nodes connected by a link, and a closed walk of length 3 is associated with a triangle. A closed walk of length 4 can visit two, three or four nodes. In Figure 4.4 the closed walks of lengths 2 to 4 are represented with their associated subgraphs.

The equivalence between closed walks and subgraphs means, in practical terms, that by counting the first ones we are able to count the second ones. Closed walks are related to the spectral moments of the adjacency matrix of the network, as we have seen in Chapter 2. Then we can relate spectral moments and the number of subgraphs of different kinds in a network. Expressions of the first spectral moments in term of subgraphs for simple networks without self-loops are:

$$\mu_0 = n \tag{4.5}$$

$$\mu_1 = 0 \tag{4.6}$$

$$\mu_2 = 2|K_2| \tag{4.7}$$

$$\mu_3 = 6|K_3| \tag{4.8}$$

$$\mu_4 = 2|K_2| + 4|P_2| + 8|C_4| \tag{4.9}$$

$$\mu_k = \sum_{i=1}^{s} c_i|S_i| \tag{4.10}$$

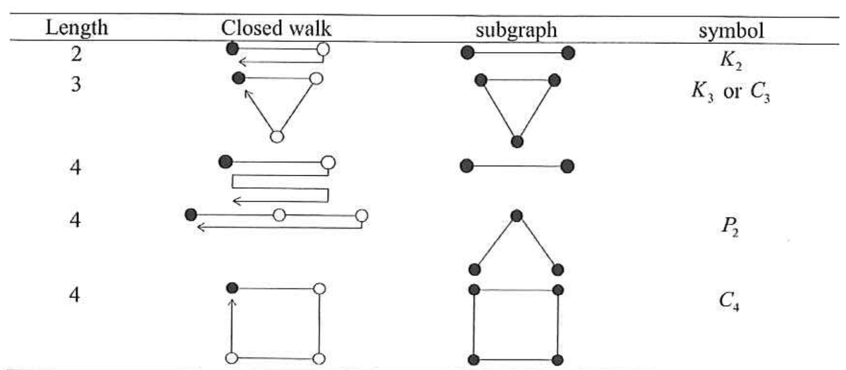

Fig. 4.4
Closed walks and subgraphs. Representation of closed walks of lengths 2 to 4, and the subgraphs associated with them.

[1] The nodes $v_1, v_2, \ldots v_l, v_{l+1}$ should be selected such as there is a link from v_i to v_{i+1} for each $i = 1, 2, \ldots, l$."

where $|S_i|$ represents the number of subgraphs of the corresponding type S_i—complete graphs K_n, paths P_n, and cycles C_n of size n, and so on. Here, C_i is an integer number representing the number of the corresponding fragment S_i. The first use of spectral moments and its 'decomposition' in terms of subgraphs dates back to the work of Jiang et al. (1984), in which an application in quantum molecular science was advanced.

4.4 Counting small undirected subgraphs analytically

Most of the methods used today for the analysis of network subgraphs are based on searching algorithms. However, it is possible to calculate analytically the number of certain small subgraphs present in a network (Alon et al., 1997). These calculations are based on the idea of spectral moments analysed in the previous section. In this section we present formulae for calculating the number of 17 small subgraphs in networks based on spectral moments of the adjacency matrix. We use the term $|S_i|$ to designate the number of subgraphs of type S_i, where the corresponding subgraph is illustrated in Figure 4.5. It is also worth noting here that some of these subgraphs correspond to known classes of graph, which are designated in other parts of this book with a different notation. For instance, the following subgraphs are cycles: $S_1 \equiv C_3$, $S_4 \equiv C_4$, $S_7 \equiv C_5$, $S_{14} \equiv C_6$; subgraph S_2 and S_3 are the path of length 3 P_3 and the star of 4 nodes $S_{1,4}$, respectively; the following subgraphs are tadpole graphs: $S_5 \equiv T_{3,1}$, $S_9 \equiv T_{4,1}$, $S_{1,5} \equiv T_{5,1}$, and $S \equiv T_{3,2}$. Here we also use the notation $\mathrm{M}^{(D)}$ for a diagonal matrix with diagonal entries $\mathbf{M}_{ii}$. The numbers of these 17 subgraphs are represented by the following expressions:

$$|S_1| = \frac{1}{6}\mu_3 \tag{4.11}$$

$$|S_2| = \frac{1}{2}\langle \mathbf{k}'|\mathbf{A}|\mathbf{k}'\rangle - 3|S_1| \tag{4.12}$$

where $\mathbf{k}' = \mathbf{k} - 1$,

$$|S_3| = \frac{1}{6}\langle \mathbf{k} \circ \mathbf{k}'|\mathbf{k}''\rangle \tag{4.13}$$

where $\circ$ denotes the Hadamard product, and $\mathbf{k}'' = \mathbf{k} - 2$,

$$|S_4| = \frac{1}{8}[\mu_4 - 4|P_2| - 2|P_1|] \tag{4.14}$$

$$|S_5| = \frac{1}{2}\langle \mathbf{t}|\mathbf{k}''\rangle \tag{4.15}$$

where $\mathbf{t} = diag(\mathbf{A}^3)$,

$$|S_6| = \frac{1}{4}\langle \mathbf{u}^T|\mathbf{Q} \circ (\mathbf{Q} - \mathbf{A})|\mathbf{1}\rangle \tag{4.16}$$

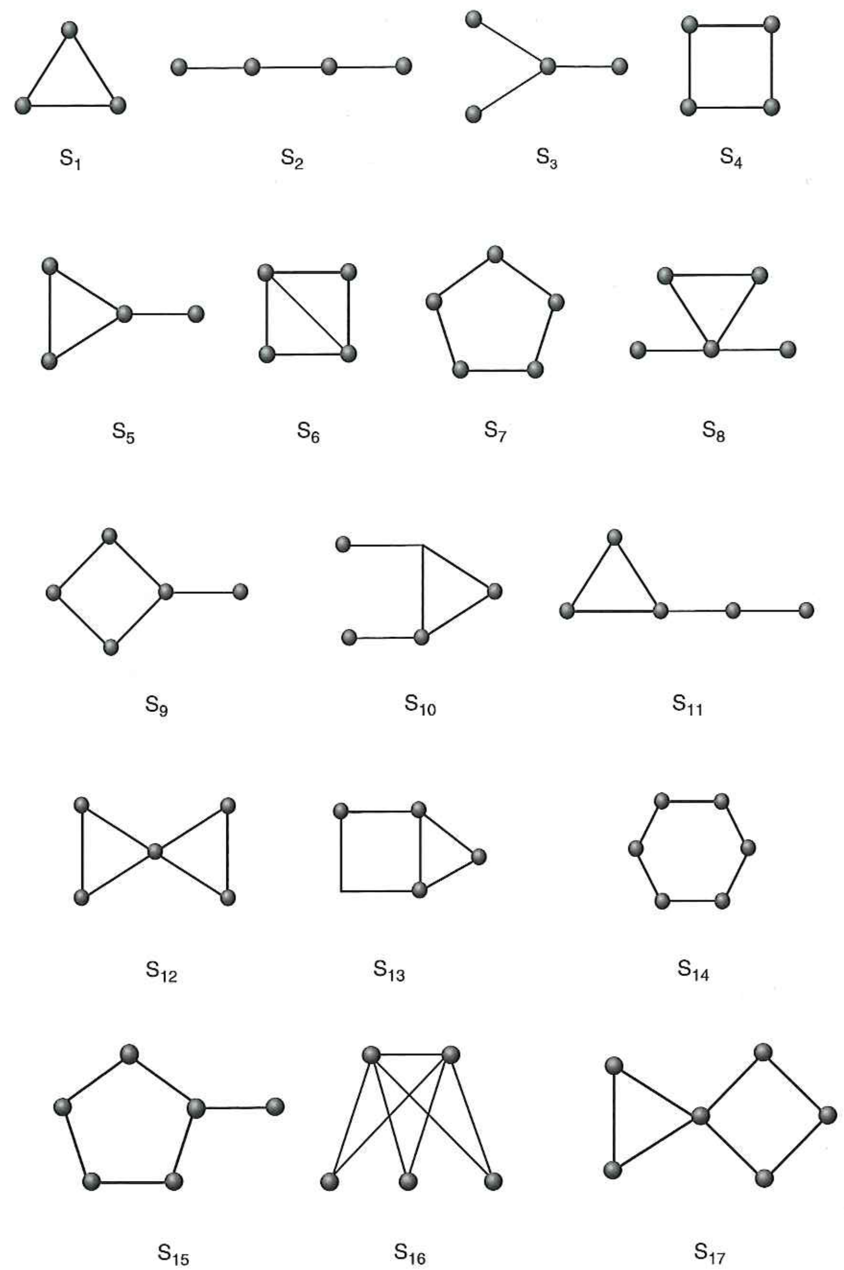

Fig. 4.5
Small subgraphs. Representation of subgraphs given by equations (4.11)–(4.27).

where $\mathbf{Q} = \mathbf{A}^2 \circ \mathbf{A}$,

$$|S_7| = \frac{1}{10}[\mu_5 - 10|S_5| - 30|S_1|] \tag{4.17}$$

$$|S_8| = \frac{1}{4}\langle \mathbf{k}'' \circ (\mathbf{k}'' - 1)|\mathbf{t}\rangle \tag{4.18}$$

$$|S_9| = \langle \mathbf{1}|\mathbf{P} - \mathbf{P}^{(D)}|\mathbf{k}''\rangle \tag{4.19}$$

where $\mathbf{P} = \frac{1}{2}[\mathbf{A}^2 \circ (\mathbf{A}^2 - 1)]$,

$$|S_{10}| = \frac{1}{2}\langle \mathbf{k}''|\mathbf{Q}|\mathbf{k}''\rangle - 2|S_6| \tag{4.20}$$

$$|S_{11}| = \frac{1}{2}\langle \mathbf{1}|\mathbf{A}^2 - \mathbf{A}^{2(D)}|\mathbf{t}\rangle \tag{4.21}$$

$$|S_{12}| = \frac{1}{4}\left\langle \mathbf{t}\left|\frac{1}{2}\,\mathbf{t} - 1\right.\right\rangle - 2|S_6| \tag{4.22}$$

$$|S_{13}| = \frac{1}{2}\langle \mathbf{1}|\mathbf{Q} \circ (\mathbf{A}^3 \circ \mathbf{A})|\mathbf{1}\rangle - 9|S_1| - 2|S_5| - 4|S_6| \tag{4.23}$$

$$|S_{14}| = \frac{1}{12}(\mu_6 - 2|P_1| - 12|P_2| - 24|S_1| - 6|S_2| - 12|S_3| - 48|S_4| - 12|S_6| - 12|S_9| - 24|S_{12}|) \tag{4.24}$$

$$|S_{15}| = \langle \mathbf{k}''|\mathbf{b}\rangle - 2|S_{13}| \tag{4.25}$$

where $\mathbf{b} = \frac{1}{2}[\mathbf{q} - 5\mathbf{t} - 2\mathbf{t} \circ \mathbf{k}'' - 2\mathbf{Q}|\mathbf{k}''\rangle - \mathbf{A}|\mathbf{t}\rangle + 2\mathbf{Q}|\mathbf{1}\rangle]$ and $\mathbf{q} = diag(\mathbf{A}^5)$,

$$|S_{16}| = \frac{1}{2}\langle \mathbf{1}|\mathbf{R} \circ \mathbf{A}|\mathbf{1}\rangle \tag{4.26}$$

where $\mathbf{R} = \frac{1}{6}\mathbf{A}^2 \circ (\mathbf{A}^2 - 1) \circ (\mathbf{A}^2 - 2)$, and

$$|S_{17}| = \frac{1}{2}\langle \mathbf{1}|\mathbf{P} - \mathbf{P}^{(D)}|\mathbf{t}\rangle - 6|S_6| - 2|S_{13}| - 6|S_{16}| \tag{4.27}$$

In Figure 4.6 we illustrate the relative abundance of the subgraphs shown in Figure 4.5 for six complex networks representing different systems in the real-world. The relative abundance is calculated as

$$\alpha_i = \frac{N_i^{real} - \langle N_i^{random}\rangle}{N_i^{real} + \langle N_i^{random}\rangle} \tag{4.28}$$

where the average is taken over random networks having the same degrees for the nodes as the real ones. The algorithm used here is the same as that used by the program MFinder, developed for identifying network motifs (Kashtan et al., 2002). Fragments 6, 12, 13, and 16 are found in these networks to be highly abundant in comparison with random networks—except for the Internet, where most of the fragments, except fragments 6 and 16, show relatively low and even negative abundance.

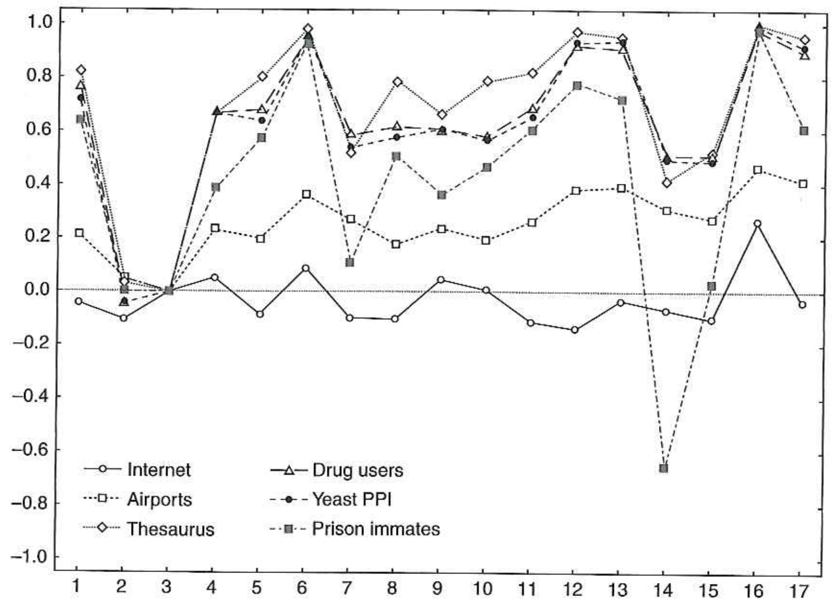

Fig. 4.6
Small subgraphs in real networks. Relative abundance of small subgraphs in six different real-world networks.

4.5 Fragment ratios

4.5.1 Clustering coefficients

In their seminal paper on 'small-world' networks, Watts and Strogatz (1998) introduced a way of quantifying the extent of clustering of a node. For a given node i, the clustering coefficient is the number of triangles connected to this node $|C_3(i)|$ divided by the number of triples centred on it:

$$C_i = \frac{2|C_3(i)|}{k_i(k_i-1)} \tag{4.29}$$

where k_i is the degree of the node. The average value of the clustering for all nodes in a network $\overline{C}$ has been extensively used in the analysis of complex networks:

$$\overline{C} = \frac{1}{n}\sum_{i=1}^{n} C_i \tag{4.30}$$

Bollobás (2003) has remarked that 'this kind of "average of an average" is often not very informative.' A second clustering coefficient was introduced by Newman (2001) as a global characterisation of network cliquishness. This index takes into account that each triangle consists of three different connected triples—one with each of the nodes at the centre. This index—also known as *network transitivity*—is defined as the ratio of three times the number of triangles divided by the number of connected triples (2-paths):[2]

$$C = \frac{3|C_3|}{|P_2|} \tag{4.31}$$

[2] Note that C is the same appearing in the assortativity coefficient, Eq. (2.20).

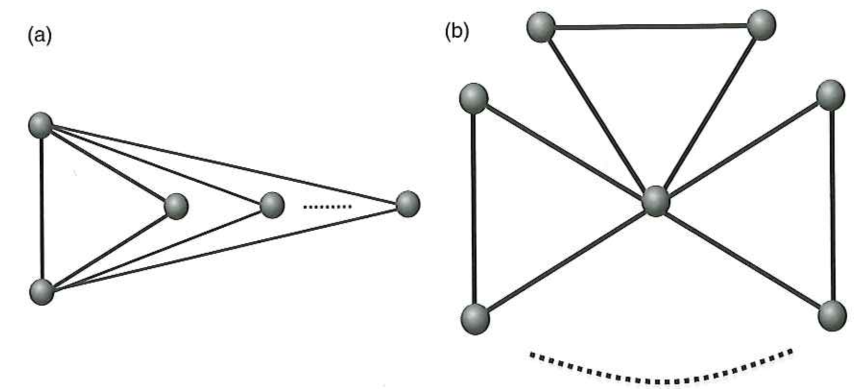

Fig. 4.7
Graphs and clustering. Two networks for which $\overline{C}$ and C diverge as the number of nodes increases.

A network with large average clustering $\overline{C}$ is not necessarily a highly clustered network, according to C. A couple of examples are provided by the networks illustrated in Figure 4.7. The first of them (A) was proposed by Bollobás (2003), and the second (B) is provided here for the first time. The graphs having the structure represented in Figure 4.7 (B) are known as 'friendship graphs' or 'Dutch-mill graphs.' The dotted lines indicate that the number of triangles in such networks can grow to infinity.

In general, we can consider that the number of triangles in these networks is designated by t. The number of nodes in these networks are $n(A) = t+2$ and $n(B) = 2t+1$. The average and global clustering coefficients are then represented by the following expressions:

$$\overline{C}(A) = \frac{4}{n(n-1)} + \frac{(n-2)}{n}, \quad \text{and} \quad \overline{C}(B) = \frac{1}{n(n-2)} + \frac{(n-1)}{n} \tag{4.32}$$

and

$$C(A) = C(B) = \frac{3}{n} \tag{4.33}$$

These expressions imply that as the number of triangles increases, the average clustering coefficient tends to 1 but the global clustering tends to zero. That is, $\overline{C} \to 1$ and $C \to 0$ as $n \to \infty$. This indicates that a network can be highly clustered at a local level but not on a global scale, and that taking the average of a local clustering does not produce a good representation of the global characteristics of the network. An important consequence of these differences in the two definitions of clustering is that the corresponding indices are not correlated to each other as expected. In Figure 4.8 we illustrate this lack of correlation between both indices for twenty real-world networks according to the data published by Newman (2003b). For practical applications, both indices are usually compared with the respective coefficients obtained for random graphs. Both indices tend to be significantly higher for real-world networks than for random networks of the same sizes and densities.

The extension of the concept of clustering as the ratio of a given subgraph to its maximum possible number has been carried out by Caldarelli et al. (2004),

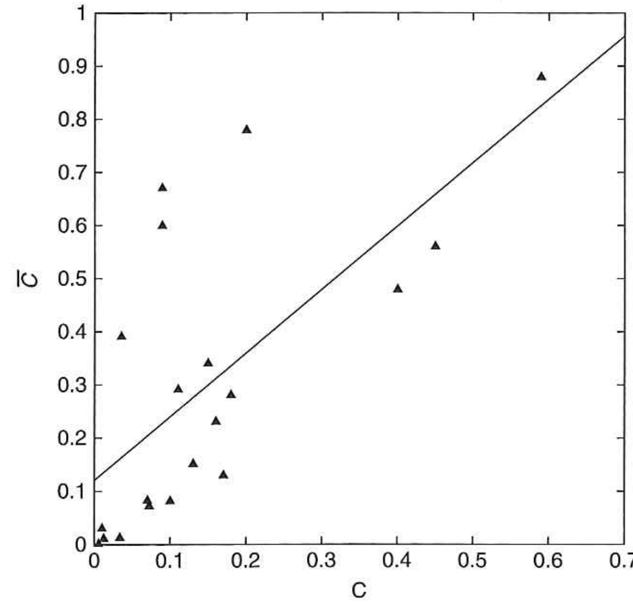

Fig. 4.8
Clustering coefficients. Correlation between the clustering coefficients defined by Watts and Strogatz (1998) and by Newman et al. (2001), $\overline{C}$ and C, respectively.

who considered the ratio of the number of different cycles in a network to its maximum possible number. In this case, the index (4.29) is just the first member of a series of analogous indices for cycles. In a similar way, Fronczak et al. (2002) have analysed the probability that there is a distance of length x between two neighbours of a node i. When the considered distance is just $x \equiv 1$, the index is reduced to the Watts–Strogatz clustering coefficient. The elimination of degree correlation biases from this coefficient has also been analysed in the literature (Soffer and Vázquez 2005).

It has been observed empirically in many real-world networks that high-degree nodes tend to display the low local clustering coefficient, C_i. That is, if we plot the average local clustering coefficient of all nodes with degree k, $C(k)$, versus degree, we obtain a clear decay, $C(k) \sim k^{-\beta}$ (Ravasz et al., 2002; Ravasz and Barabási, 2003). Such a kind of relationship is not observed for random networks with Poisson or power-law degree distributions (Ravasz and Barabási, 2003). The presence of this power-law decay of the clustering with the degree has been attributed to the existence of particular 'hierarchical' structure in complex networks. A hierarchical network according to Ravasz et al. (2002) is 'one made of numerous small, highly integrated modules, which are assembled into larger ones.' Such structure is illustrated in Figure 4.9.

As we have already seen for some graphs like those depicted in Figure 4.7, a network can display a very large average clustering coefficient, despite the possibility of the highest-degree nodes having clustering close to zero. Let us, for instance, consider a star-like node i with degree k_i. If this node has clustering coefficient C_i, the number of triangles in which it take place is represented by

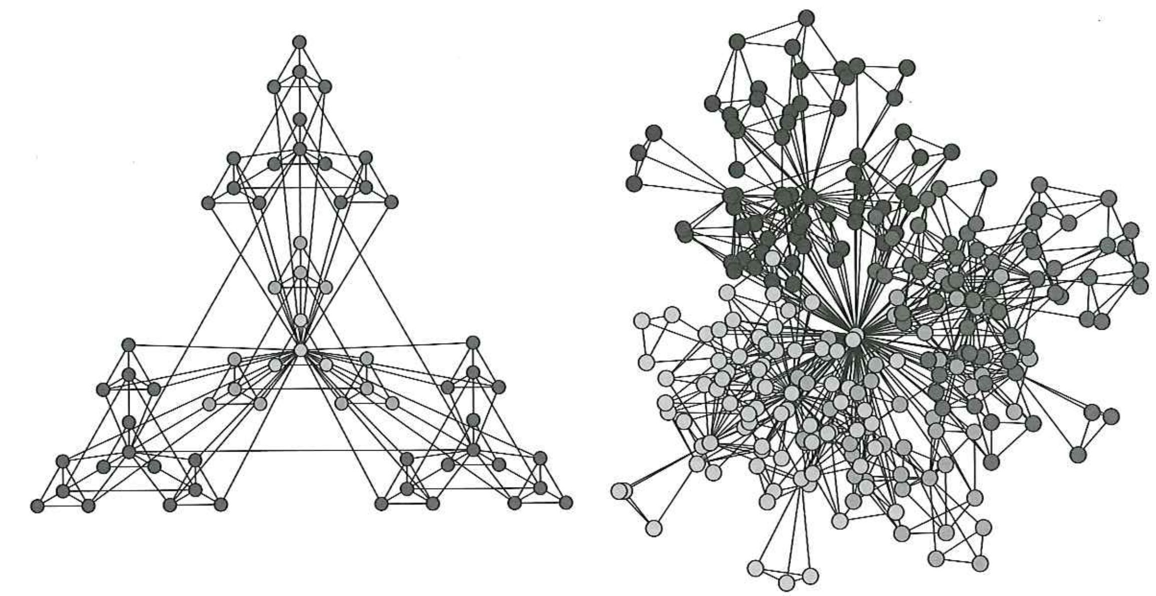

Fig. 4.9
Hierarchical networks. The hierarchical networks which have scale-free topology with embedded modularity. From Ravasz, E., Somera, A. L. Mongru, D. A., Oltvai, Z. N., and Barabási, A.-L. (2002). 'Hierarchical organization of modularity in metabolic networks'. *Science*, **297**, 1551–5. (Reproduced with the permission of AAAS.)

$$\Delta_i = \frac{1}{2} C_i k_i (k_i - 1) \tag{4.34}$$

The most efficient way of increasing the number of triangles in which node i takes place is to create cliques of size l having i as one of its nodes. For instance, in Figure 4.10 we illustrate this situation for a star graph of five nodes (A). In (B) and (C), two graphs are shown in which the central node i has clustering coefficient $C_i = 2/3$. However, in network (C), where the central node takes part in a 4-clique, the number of links is smaller than in (B). Therefore, we consider B to be a more 'efficient' way of increasing the clustering of a node, as it requires the creation of a smaller number of links in the graph.

The number of triangles in an l -clique is represented by

$$\Delta_l = \frac{1}{6} l (l^2 - 1) \tag{4.35}$$

Then, the number of l-cliques in which node i with degree k_i and clustering coefficient C_i can take place is

$$NC_l = \left[\frac{\Delta_i - \left(\sum_{j=1}^{l-3} NC_{l-j} \cdot \Delta_{l-j} \right)}{\Delta_l} \right] \tag{4.36}$$

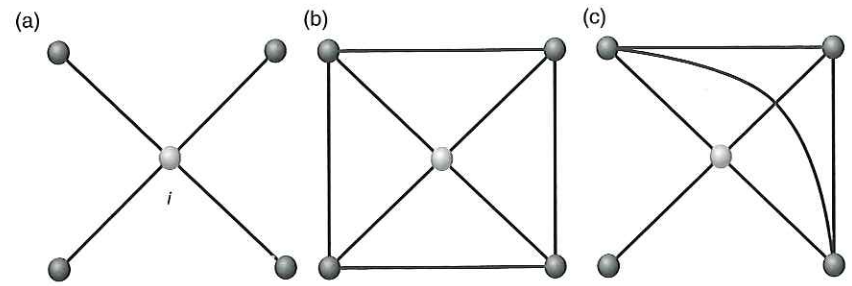

Fig. 4.10
Local clustering. A star graph of five nodes, at left, and two ways of increasing the local clustering of node i. Those at centre and right have $C_i = 2/3$, but in the latter the number of links added to the one at left is smaller than that in the centre.

For instance, let us consider a node i with degree $k_i = 8$. If $C_i = 0.357$, the number of triangles in which it takes place is equal to $\Delta_i = 10$. This node can be part of an l -clique where $3 \leq l \leq 8 = k_i$. Then, applying (4.36), we have

$$NC_8 = \left\lfloor \frac{10}{56} \right\rfloor = 0 \tag{4.37}$$

$$NC_7 = \left\lfloor \frac{10 - 0{\cdot}56}{35} \right\rfloor = 0 \tag{4.38}$$

$$NC_6 = \left\lfloor \frac{10 - 0{\cdot}56 - 0{\cdot}35}{20} \right\rfloor = 0 \tag{4.39}$$

$$NC_5 = \left\lfloor \frac{10 - 0{\cdot}56 - 0{\cdot}35 - 0{\cdot}20}{10} \right\rfloor = 1 \tag{4.40}$$

$$NC_4 = \left\lfloor \frac{10 - 0{\cdot}56 - 0{\cdot}35 - 0{\cdot}20 - 1{\cdot}10}{4} \right\rfloor = 0 \tag{4.41}$$

$$NC_3 = \left\lfloor \frac{10 - 0{\cdot}56 - 0{\cdot}35 - 0{\cdot}20 - 1{\cdot}10 - 0{\cdot}4}{1} \right\rfloor = 0 \tag{4.42}$$

That is, for a node i with degree $k_i = 8$ in order to have $C_i = 0.357$, it is enough to have a 5-clique. However, for the same node to have $C_i = 0.928$, the same calculations show that it is necessary to have one 6-clique, one 4-clique, and two 3-cliques. If we consider a node with degree $k_i = 20$, it is necessary to embed it into one 11-clique, one 5-clique, and one 3-clique in order to have a clustering coefficient $C_i = 0.926$. In addition, for a node of degree $k_i = 100$ to have $C_i \approx 0.9$, it is necessary to embed it into a clique of 31 nodes. This means that as the degree of a node increases, the sizes of the cliques in which it needs to take place in order to maintain the same clustering, increases dramatically. The creation of such huge cliques is very costly in terms of the number of links that need to be added around a node, which dramatically increases the density of the network. Then, it is expected that high-degree nodes display very low clustering as a consequence of their participation in small cliques only. Consequently, in a network for which the number of cliques of given sizes decays very fast with the size of these cliques, a power-law decay of the clustering with the degree is expected as a consequence of the previous

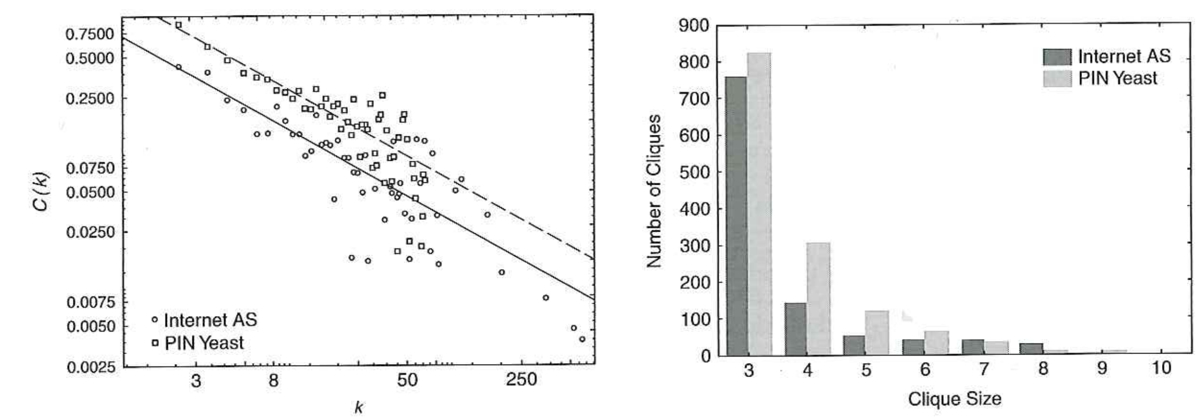

Fig. 4.11
Local clustering and node degree. Plot of the local mean clustering coefficient averaged over all nodes with the same degree (left) for two real-world networks. Histogram of the distribution of number of cliques in terms of clique size (right).

analysis. Of course, modular hierarchical networks are only one example of this kind of structure.

In Figure 4.11 we illustrate the plot of $C(k)$ versus k for two real networks representing the Internet as an autonomous system and the PIN of yeast. In both cases, the power-law correlation obtained is of the form $C(k) \sim k^{-0.69}$. That is, in both cases the nodes with the highest degree display very low clustering, and those with low degree are relatively very clustered. This version of the Internet has 17 nodes with a degree higher than 50, including nodes with degree 590, 524, 355, and so on. The PIN of yeast has 14 nodes with degree higher than 50. Thus, for these nodes to display large clustering it is necessary to have very large cliques in these networks. However, as can be seen in Fig 4.11 (B), none of these networks have cliques with more than 9 nodes, and the number of cliques decreases very quickly with the size of the clique. Thus, the existence of the relations observed in Figure 4.11 (A) is a direct consequence of the absence of large cliques in these networks more than the presence of any hierarchy in the modules of the network.

4.5.2 Network reciprocity

Another index which is based on the ratio of the number of certain subgraphs in a network is the reciprocity, defined for a directed network as

$$r = \frac{L^{\leftrightarrow}}{L} \tag{4.43}$$

where $L^{\leftrightarrow}$ is the number of symmetric pairs of arcs, and L is the total number of directed links (Wasserman and Faust, 1994; Newman et al., 2002; Serrano and Boguñá, 2003). Thus, if there is a link pointing from A to B, the reciprocity measures the probability that there is also a link pointing from B to A.

Table 4.1 **Network reciprocity.** Some reciprocal and antireciprocal networks.

Reciprocal	ρ	Areciprocal	ρ	Antireciprocal	ρ
Metab. *E. coli*	0.764	Transc. *E. coli*	–0.0003	Skipwith	–0.280
Thesaurus	0.559	Transc. yeast	$-5.5.10^{-4}$	St. Martin	–0.130
ODLIS	0.203	Internet	$-5.6.10^{-4}$	Chesapeake	–0.057
Neurons	0.158			Little Rock	–0.025

A modification of this index was introduced by Garlaschelli and Lofredo 2004a), who considered the following normalised index:

$$\rho = \frac{r - \bar{a}}{1 - \bar{a}} \tag{4.44}$$

where $\bar{a} = L/n(n-1)$. In this case, the number of self-loops are not considered for calculating L, such as $L = \sum_{i=1}^{n}\sum_{j=1}^{n} A_{ij} - \sum_{i=1}^{n} A_{ii}$, where A_{ij} is the corresponding entry of the adjacency matrix $\mathbf{A}$ of the directed network. It is also straightforward to realise that $L^{\leftrightarrow} = tr(\mathbf{A}^2)$. According to this index, a directed network can be reciprocal ($\rho > 0$), areciprocal ($\rho = 0$), or antireciprocal ($\rho < 0$). Some examples of network reciprocity are provided in Table 4.1.

Accounting for all parts (subgraphs)

5

> ...the different parts of each being must be co-ordinated in such a way as to render the existence of the being as a whole, not only in itself, but also in its relations with other beings, and the analysis of these conditions often leads to general laws which are as certain as those which are derived from calculation or from experiment.
>
> Georges Cuvier

We have seen in previous sections that small subgraphs form the basis of important structural characteristics of networks, such as motifs, fingerprints and clustering coefficients. The equivalence between spectral moments and subgraphs allows the definition of new topological invariants to describe the structure of a network in terms of subgraphs. For instance, let us consider a simple function based on the first spectral moments:

$$f(G, c) = \sum_{k=0}^{4} c_k M_k \tag{5.1}$$

This invariant represents a weighted sum of the numbers of the smallest subgraphs in the network:

$$f(G, c) = c_0 n + 2(c_2 + c_4)|K_2| + 4|P_2| + 6c_3|C_3| + 8|C_4| \tag{5.2}$$

It is straightforward to generalize this kind of invariant to account for the presence of any subgraph in the network. This kind of invariant is based on the infinite sum of all spectral moments of the adjacency matrix:

$$f(G, c) = \sum_{k=0}^{\infty} c_k M_k \tag{5.3}$$

The key element in the definition of invariants based on the expression (5.3) is the selection of the coefficients c_k. It is known that the infinite sum of spectral moments in the network diverges: $\sum_{k=0}^{\infty} M_k = \infty$. Consequently, we need to select the coefficients c_k in such a way that the series (5.3) converges. As we require small subgraphs to make larger contributions than big ones, the coefficients c_k can be selected in such a way that the closed walks of small